PARSEC 7.0 – ANNUAL TECHNICAL FESTIVAL
INDIAN INSTITUTE OF TECHNOLOGY DHARWAD

VENI VIDI VICI

Think Like an Attacker. Reason Like a Defender.

CYBERSECURITY CAPTURE THE FLAG COMPETITION
Organized by: **Coding Club, IIT Dharwad**

1. Introduction

PARSEC, the annual technical festival of the Indian Institute of Technology Dharwad, brings together students, innovators, and technology enthusiasts to explore emerging technologies and demonstrate their technical abilities.

As part of PARSEC 7.0, **The Systems Wing** of the Coding Club, Indian Institute of Technology Dharwad, proudly presents **Veni Vidi Vici**, a cybersecurity Capture The Flag (CTF) competition designed to challenge participants across multiple domains of information security.

Veni Vidi Vici aims to go beyond conventional cybersecurity competitions by introducing participants to **realistic attack scenarios**, **carefully designed vulnerable environments**, and **multi-stage challenges** that demand analytical thinking, technical proficiency, and creative problem-solving.

The competition follows a scenario-driven approach, **encouraging participants to think like attackers while understanding the underlying principles of defensive security.** From network investigations and application vulnerabilities to cloud misconfigurations, cryptographic weaknesses, and emerging AI security threats, participants will encounter challenges inspired by real-world cybersecurity incidents.

Through an immersive and hands-on experience, **Veni Vidi Vici** aspires to foster practical cybersecurity skills, encourage security research, and create an environment where students can explore the evolving landscape of digital security.

2. About the Organizers

The Systems Wing of the Coding Club of IIT Dharwad is a student-driven technical community dedicated to promoting systems programming, software development, hardware integrations, embedded systems and emerging areas of computer science through collaborative learning, technical projects, workshops, and competitive events.

The wing actively encourages students to explore domains such as:

- Systems programming and low-level computing.
- Artificial intelligence and machine learning.
- Cybersecurity and information security.
- Hardware and embedded systems based integrations.
- Distributed systems, cloud computing, and networking.

Through Veni Vidi Vici, the club seeks to extend its technical initiatives into practical cybersecurity education by providing participants with an opportunity to engage with realistic security problems in a controlled environment.

3. Proposed Event Structure

Veni Vidi Vici: Capture The Flag Competition

The competition will feature a collection of carefully curated cybersecurity challenges distributed across six major domains.

The proposed challenge architecture consists of **12 primary challenge environments**, with approximately **24–30 flags** distributed across different stages of investigation and exploitation.

Challenges will be organized across three progressive difficulty levels:

- **Foundation:** Fundamental cybersecurity concepts and introductory problem-solving.
- **Intermediate:** Multi-step investigations requiring practical technical knowledge.
- **Advanced:** Complex attack scenarios requiring deeper analysis, exploitation, and lateral reasoning.

Challenge Domains

1. Network Security

Challenges involving packet analysis, network reconnaissance, traffic investigation, protocol vulnerabilities, and digital forensics.

2. Application Security

Exploration of common web application vulnerabilities, authentication weaknesses, insecure APIs, injection attacks, and access-control failures.

3. Cloud Security

Challenges involving cloud infrastructure misconfigurations, identity and access management, exposed resources, and security weaknesses in cloud environments.

4. System Security and Privilege Escalation

Investigation of vulnerable systems, permission misconfigurations, privilege escalation techniques, and operating-system security.

5. Applied Cryptography

Problems involving cryptographic implementation weaknesses, encryption and encoding mechanisms, cryptanalysis, and secure communication.

6. AI Systems Security

Exploration of emerging vulnerabilities in AI-powered systems, including prompt injection, insecure agent interactions, and weaknesses in AI-integrated applications.

Challenge Design Philosophy

The competition emphasizes realistic, scenario-driven challenges rather than isolated technical exercises.

Participants will encounter interconnected investigation stages that may involve:

- Reconnaissance and information gathering.
- Initial access and vulnerability identification.
- Enumeration and forensic investigation.
- Credential discovery and exploitation.
- Privilege escalation and lateral movement.
- Achievement of the final objective.

The challenge environments will be sandboxed and designed to provide a controlled setting for practical cybersecurity experimentation.

Technical Resources and Learning Opportunities

Alongside the competition, Veni Vidi Vici will feature **learning resources** aimed at introducing participants to selected cybersecurity domains.

These resources will aim to:

- Introduce fundamental concepts in different cybersecurity domains.
- Familiarize participants with commonly used security tools.
- Bridge the gap between theoretical understanding and hands-on cybersecurity practice.

The resources will be designed to accommodate participants with varying levels of prior experience.

4. Collaboration and Industry Involvement

We invite organizations, cybersecurity companies, security researchers, and industry professionals to collaborate with Veni Vidi Vici and contribute to the development of practical cybersecurity education at IIT Dharwad.

Possible forms of collaboration include:

- **Challenge Partnerships:** Contributing real-world security scenarios, vulnerabilities, or problem statements that can be adapted into competition challenges.
- **Technical Mentorship:** Providing cybersecurity professionals and researchers to mentor participants and guide technical sessions.
- **Knowledge Partnerships:** Collaborating on workshops, demonstrations, and discussions covering emerging areas of cybersecurity.
- **Judging and Evaluation:** Participating in the evaluation of challenges, technical presentations, and competition outcomes.
- **Infrastructure Partnerships:** Supporting the event through cybersecurity platforms, cloud infrastructure, software licenses, and technical resources.
- **Prize and Sponsorship Partnerships:** Supporting participant incentives, awards, and event execution.

Industry participation will provide organizations with an opportunity to engage with technically motivated students, demonstrate their expertise, and contribute to strengthening the cybersecurity ecosystem.

We also welcome opportunities to adapt industry-relevant security scenarios into educational challenge environments, ensuring that participants gain exposure to problems inspired by practical security research and professional workflows.

5. Tentative Timeline

Phase	Tentative Dates	Description
Event Announcement	[19th December, 2026]	Announcement of Veni Vidi Vici and opening of registrations.
Registration	[20th December, 2026 – 24th January, 2027]	Participant registration and team formation.
Release of CTF Materials	[20th January, 2027]	Release of competition relevant materials, literature and resources.
Competition Launch	[25th January, 2027]	Commencement of the CTF competition and release of challenges.
Competition Conclusion	[27th January, 2027]	Closure of the competition and final evaluation.
Results and Awards	[29th January, 2027]	Announcement of results and recognition of winning participants.

The final schedule and competition format will be announced closer to the event.

6. Event Vision

Veni Vidi Vici aspires to create more than a conventional cybersecurity competition. It seeks to establish an environment where curiosity meets practical security research, where theoretical knowledge is tested against realistic challenges, and where participants develop the ability to understand and reason about complex security problems.

By bringing together students, cybersecurity professionals, and industry organizations, the event aims to promote awareness of emerging security threats, encourage technical exploration, and nurture the next generation of cybersecurity practitioners.

We warmly welcome organizations and professionals working in cybersecurity, information security, cloud security, application security, and related domains to collaborate with us in making Veni Vidi Vici a meaningful technical experience.

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